Adversarial examples

AI security · University · Lecture 1



In this lecture

- Explain why adversarial examples exist.
- Compute and run FGSM.
- Evaluate defences honestly.

Remember

- Small changes add up.
- Train on attacks.
- Attack the defence.

Attack, defend, attack again

1. Warm-up (5 min): What does the model see?
2. Attack (15 min): By hand.
3. Defend (15 min): Propose.
4. Break (20 min): Adapt.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which defences survived?
2. What did the threat model leave out?
3. How would you report robustness?

Exit ticket

- Name one claim you will no longer accept without an adaptive attack.